BT-403 MIDTERM- 100% GUESS PAPER 2023

Agricultural Biotechnology

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OBJECTIVES

1. Agricultural biotechnology is_____Collection of scientific techniques, To improve plants, animals and microorganisms and production or processing of agricultural products, Manipulate the genetic makeup, ***All the above given***
2. Which hormone is abundantly available in plants? ***Auxin***
3. Totipotent plant cells are those which have the ability to grow in ***culture or in vitro conditions.***
4. A ***germplasm*** is a collection of genetic resources for an organism.
5. Markers with visual identification are called? ***Morphological Markers***
6. The genetic markers should be_____and_____: ***palymorphic;. heterazygous***:
7. Plastid genome is_____ ***60 copies of a single circular chromosomes per plastid and 50-60 chloroplast per cell***
8. Resistant to two or more herbicide families with same mechanism of action is known as_____ ***Cross Herbicide Resistance***
9. ***Morphological*** markers are those traits that are scored visually, or morphological markers are those genetic markers whose inheritance can be followed with the naked eye.
10. ***Tissue culture*** the process of growing cells artificially in the laboratory.
11. The development of shoot from the callus is called____ ***caulogenesis***
12. Isolation of tissue is immersed ***1.0*** molar sucrose.
13. FEG used? ***Ca2+***
14. Markers with visual identification are called? ***Morphological Markers***
15. PEG (Polyethylene glycol) is used as a ***medium or agent*** for cell fusion or transformation in certain biological and biotechnological applications.
16. The culture medium with the inoculum is incubated at 26 - 28 C with the light intensity at ***2000-4000*** lux (unit of intensity of light) and allowing photoperiod of 16 hour of light and 8 hours of darkness.
17. Causes of Somaclonal Variations are? Biochemical, physical, chemical, ***All of them***
18. ***Indirect organagenesis*** process is more important for transgenic plant production.
19. The pH of the medium is adjusted to___ ***5.8***
20. ***64%*** of total soya crop
21. Transformation was first demonstrated in **1928** by British bacteriologist ***Frederick Griffith.***
22. Griffith discovered that a harmless strain of ***Streptococcus pneumoniae***
23. Herbicide type ***systematic type***
24. In which the culture can be prepared… ***all above***
25. Drought is …..***Stomata closure***
26. Anthers or pollens can be cultured on a suitable medium containing sucrose ***(usually 2%)***, iron, vitamins, hormones.

27. A ***2007*** systematic review found that "most studies on non-Hodgkin lymphoma and ***leukemia*** showed positive associations with pesticide exposure"
28. Indirect organogenesis used for ***callus phase*** transgenic plant.
29. Surface sterilize with ***0.01% Tween-20 for 15 min***
30. The process of growing an entire plant from a single cell or group of cells. ***Regeneration***
31. ***Transposable elements*** (TEs), also known as "jumping genes" or ***transposons,***
32. particles of gold or tungsten ***(0.6 – 1 mm in size).***
33. Photosynthesis is carried out in presence if which plastid….***chloroplast***
34. Growth is change in ……***size***
35. ***Ovicides*** are used to control eggs of insects and mites.

SUBJECTIVE

Define Pesticides:

Pesticides are chemical substances used to control or eliminate pests, such as insects, weeds, and fungi, in order to protect crops, livestock, and public health from their detrimental effects.

Mechanism of resistant in plants, tissue culture:

Resistant mechanisms in plants involve genetic adaptations that enable them to withstand the harmful effects of pests or diseases. These mechanisms can include physical barriers, production of toxic compounds, and altered biochemical pathways. This enables the production of disease-free and genetically identical plants, providing a valuable tool for breeding. Plants can also produce toxic compounds, like alkaloids or terpenes, that deter pests or inhibit pathogen growth. Another mechanism involves the activation of defense responses, such as the production of antimicrobial proteins or phytohormones, to fight off invaders. Additionally, plants can undergo genetic changes, including the activation of resistance genes or the modification of metabolic pathways, to enhance their resistance against specific threats.

Tissue culture:

Tissue culture is a laboratory technique that involves the propagation of plant cells, tissues, or organs in a controlled environment to produce genetically identical and disease-free plants for various applications.

Application of DNA based marker:

DNA-based markers have a wide range of applications across different fields. In agriculture, they are used for marker-assisted breeding, genetic mapping, and identifying desirable traits in crops. In medicine, they assist in diagnosing genetic disorders, determining drug responses, and predicting disease risks. In forensic science, DNA markers aid in criminal investigations and identifying individuals. They are also utilized in evolutionary studies, conservation genetics, and studying population dynamics. Additionally, DNA markers play a role in biodiversity assessment, paternity testing, and tracking the spread of infectious diseases.

Germplasm preservation:

Germplasm is living tissue from which new plants can be grown. It can be a seed or another plant part a leaf, a piece of stem, pollen or even just a few cells that can be turned into a whole plant. Germplasm preservation refers to the collection, storage, and maintenance of plant or animal genetic resources to ensure their long-term availability for future use in breeding, research, and conservation purposes.

How Pesticides more effects kids than adults:

Kids' bodies are smaller and still growing, they take more breaths per minute, and they also eat and drink more relative to their weight. They can easily become exposed to pesticides (via inhalation or skin contact) in schools, daycare, playgrounds, hospitals. Their little kidneys and liver also cannot eliminate pesticides from their bloodstream as effectively as an adult's. all factors that make them more likely to absorb pesticides and residue.

Why plastid is used for transformation.

Bacteria can take up foreign DNA in a process called transformation. Transformation is a key step in DNA cloning. Plastids, particularly chloroplasts, are used for transformation in plants because they contain their own DNA and can be genetically modified to express desired traits. This allows for targeted modifications in photosynthesis, metabolic pathways, and other important cellular functions.

What are herbicides?

A herbicide is a chemical substance used to control or manipulate undesirable vegetation, especially weeds. Herbicides are extensively used in gardening, farming, and landscape turf management.

Pre-zygotic barriers and how to overcome :

Pre-zygotic barriers in agriculture refer to reproductive mechanisms that prevent or hinder successful fertilization between different plant varieties or species. Examples include differences in flowering time, incompatible pollinators, or physical barriers. This Can be overcome by 1= In-vitro fertilization 2= Protoplast fusion 3= Embryo culture 4= Use of growth hormones (IAA, NAA) 5= Adopting bridging species technique.

Explain indirect organogenesis:

Indirect organogenesis is a process in tissue culture where new plant organs, such as shoots or roots, are regenerated from undifferentiated callus tissue or through intermediate stages, such as somatic embryos, before developing into complete plantlets.

Limitation of PEG:

1. The DNA is susceptible for degradation and rearrangement. 2. Random integration of foreign DNA into genome may result in undesirable traits. 3. Regeneration of plants from transformed protoplasts is a difficult task.

Surface Sterilization of Explants:

Surface sterilization of explants is a critical procedure in tissue culture. It involves removing potential contaminants from the surface of plant material to establish a sterile culture. Typically, the explants are washed with a sterilizing agent like sodium hypochlorite (bleach) or ethanol to eliminate bacteria, fungi, and other microorganisms. The concentration and duration of sterilization depend on the plant species and explant type. Proper surface sterilization ensures aseptic conditions, minimizing the risk of contamination and promoting successful establishment and growth of explants in culture.

Mode of action of herbicide:

1= A herbicide's effectiveness is strongly influenced by its toxic mode of action and the application method. 2= Herbicides can act by inhibiting a plant's amino acid production, growth, photosynthesis, cell division, or by mimicking natural auxin hormones to cause deformities. 3= Most modern herbicides are synthetic mimics of a natural plant's hormones that obstruct the target plant's growth. Some plants such as the tree of heaven and juglans (walnuts) produce their own natural herbicide. 4= Organic herbicides are useful and are commonly used in organic gardens, but they are less effective and more costly than synthetic herbicides because they based on natural materials.

Define genetic markers? Also write its types and mostly used genetic markers.

A genetic marker is a gene or DNA sequence with a known location on a chromosome. It can be used to identify individuals or species. It can be described as a variation that can be observed.

Types of genetic markers:

1= Molecular markers: A sequence of DNA or protein that can be screened to reveal key attributes of its state or composition and thus used to reveal genetic variation.

2= Morphological markers: Morphological markers are those traits that are scored visually, or morphological markers are those genetic markers whose inheritance can be followed with the naked eye.

Mostly used genetic markers:

Molecular markers are the most commonly used in agricultural research due to their versatility and precision.

Morphogenesis:

Formation of new organs from the callus under the influence of auxin and cytokinin is called morphogenesis. Roots and shoots are differentiated from the callus. Such embryos are called somatic embryos result in the formation of young plantlet.

Define callus:

The callus is an unorganized mass of undifferentiated tissue. The mechanism of callus formation is that auxin induce cell elongation and cytokinin induces cell division as a result of which masses of cells are formed.

Effect of pesticides on human:

Experts broadly classify the effects of pesticides as topical or systemic. Topical reactions are usually limited to areas of the body that have come in direct contact with a pesticide. Inflammation of the skin (dermatitis) such as a rash or blisters is usually the most common topical symptom. Pesticides can have adverse effects on human health, including acute poisoning, chronic exposure leading to long-term health issues, developmental and reproductive problems, neurological disorders, and an increased risk of certain cancers.

What are environmental effects of pesticides?

Widespread use of pesticides in agriculture has experts worried due to their long term environmental damage. Environmental effects of pesticides include ecotoxicity, water contamination, soil degradation, air pollution, disruption of natural ecosystems, and impacts on beneficial insects, birds, and aquatic organisms.

Advantages of using herbicides:

1= Broad spectrum of weeds controlled, 2= Reduced crop injury, 3= Reduced herbicide carryover, 4= New mode of action for resistance management, 5= Crop management flexibility and simplicity, 6= Use of herbicides that are more environmentally friendly.

Short-term effects of pesticides:

1= Nausea and vomiting 2= Diarrhea 3= Loss of consciousness 4= Seizures 5= Coughing and sore throat 6= Extreme weakness.

Steps of pesticides:

The steps involved in pesticide use typically include: 1) identifying the pest problem, 2) selecting appropriate pesticides, 3) applying them according to label instructions, 4) monitoring effectiveness, and 5) taking necessary safety precautions to minimize environmental and human health impacts.

Steps of tissue culturing:

1= selection of explant, 2= Conti, 3= Establishment of the explant, 4= Multiplication, 5= Root formation.

Which media is used in tissue culture?

The important media used for all purpose experiment are Murashige and Skoog medium (MS medium). Others: Media commonly used in tissue culture include Murashige and Skoog (MS) medium, Gamborg's B5 medium, and Woody Plant Medium (WPM).

Why we use synthetic seeds:/ Advantages Of Synthetic Seeds

Synthetic seeds are used for their advantages in convenient handling, mass production, disease-free propagation, long-term storage, and efficient transfer of desired traits. They offer a practical alternative to conventional seeds and facilitate clonal propagation and preservation of valuable plant genetic resources.

Haploid cells:

Haploid cells are cells that contain a single set of chromosomes, representing half the normal chromosome complement of a species. They are produced through processes such as meiosis or in vitro techniques and have unique applications in genetics and plant breeding.

Why do we need to check colonies?

Several colonies are checked to identify one with the right plasmid (e.g., by PCR or restriction digest).The bacteria that make colonies should all contain a plasmid (which provides antibiotic resistance). However, it’s not necessarily the case that all of the plasmid-containing colonies will have the same plasmid.

Mechanism of Somaclonal Variations:

Genetic (Heritable Variations): 1= Pre-existing variations in the somatic cells of explant 2= Caused by mutations and other DNA changes 3= Occur at high frequency.

Epigenetic (Non-heritable Variations): 1= Variations generated during tissue culture 2= Caused by temporary phenotypic changes 3= Occur at low frequency.

Benefits of agriculture biotechnology:

1. Increased crop yields and productivity.
2. Enhanced resistance to pests, diseases, and environmental stresses.
3. Improved nutritional content of crops.
4. Efficient use of resources, such as water and fertilizers.
5. Reduction in chemical pesticide use.
6. Preservation of biodiversity through disease and pest management.
7. Development of crops with enhanced quality traits.
8. Accelerated breeding processes for crop improvement.
9. Contribution to sustainable agriculture and food security.
10. Potential for addressing global challenges like climate change and population growth.

Limitations of particle bombardment method :

1= Limited regeneration capacity of tissue being bombarded 2= Efficiency of stable integration of DNA. 3= Insertion of multiple copies of the gene 4= Integration of rearranged and/or truncated DNA sequences 5= Damage to the cellular tissue. 6= Specialized and expensive equipments are required.

Steps of micro propagation:

Micropropagation is a complicated process and comprises mainly 4 stages (I, II, III and IV). But initial step 0 is also necessary.

- Step 0:
- Stage I – Establishment:
- Stage II – Multiplication:

- Stage III – Rooting:
- Stage IV – Acclimatization:

Advantages of biochemical marker:

1= Simple 2= Inexpensive 3= Electrophoretically resolvable, and detectable 4= Does not require DNA extraction or the availability of sequence information, primers or probes 5= Quick and easy to use.

Symptoms of peripheral neuropathy:

Symptoms of peripheral neuropathy can include numbness, tingling, pain, weakness, loss of coordination, and sensitivity to touch or temperature changes in the affected areas.

Multi-Allelic viruses:

Multi-allelic viruses include Influenza A virus, Hepatitis C virus, Human immunodeficiency virus (HIV), and Papillomavirus, among others.

Multiplication step of tissue culture:

The multiplication step in tissue culture involves the proliferation and growth of cells, tissues, or organs under controlled conditions to produce an increased number of viable and healthy plantlets.

Advantages of using herbicides:

1= Broad spectrum of weeds controlled 2= Reduced crop injury 3= Reduced herbicide carryover 4= New mode of action for resistance management 5= Crop management flexibility and simplicity 6= Use of herbicides that are more environmentally friendly.

Disadvantages of herbicides:

1= Mammalian toxicity 2= Eco toxicity 3= Weeds become super weeds 4= Reduced crop yield 5= Creates soil and air pollution Herbicides also damage the Crop plants along with weed.

Who activated charcoal effects on anther and pollen culture ?

The effects of activated charcoal on anther and pollen culture are due to its ability to adsorb inhibitory substances, promote callus induction, enhance shoot regeneration, and provide a suitable microenvironment for growth and development.

Process of development of plant:

The process of plant development involves germination of seeds, root and shoot formation, leaf growth, flower initiation, pollination, fertilization, fruit development, and ultimately seed dispersal, completing the life cycle.

Disadvantages of particle bombardment:

Some disadvantages of particle bombardment (biolistics) include low transformation efficiency, random integration of foreign DNA, potential for DNA damage, limited target cell availability, and high cost and technical complexity.

Characteristics of growth of cell suspension:

Some characteristics of the growth of cell suspension cultures include rapid cell proliferation, dependency on continuous agitation for uniform nutrient distribution, requirement for a balanced nutrient medium, need for adequate oxygen supply, and the potential for producing valuable secondary metabolites.

Role of auxin in plants:

Auxin plays a crucial role in plants, promoting cell elongation, root development, apical dominance, tropic responses, and differentiation of vascular tissues, as well as influencing fruit development and abscission.

Name of chloroplast method:

The name of the method used for isolating chloroplasts from plant cells is known as "chloroplast isolation" or "chloroplast enrichment" method.

What is treatment of skin burned:

The treatment of skin burns primarily involves immediate first aid measures such as rinsing the affected area with cool water, applying sterile dressings, seeking medical attention, and following appropriate medical treatment protocols for burns.

What is crown mady disease:

Crown gall disease, also known as crown mady disease, is a plant disease caused by the bacterium Agrobacterium tumefaciens. It leads to the formation of tumor-like growths on the crown and roots of infected plants, negatively impacting their growth and productivity.

Write five name of tissue culture in plant:

1= Shoot culture 2= Callus culture 3= Embryo culture 4= Root culture 5= Anther culture.

What are the optimal conditions for protoplast culture:

1= Optimal density to the culture. 2= Optimal auxin to cytokinin ratio, glucose and sucrose. 3= Maintain osmoprotectant in the medium 4= Temperature: 20-28°C pH: 5.5-5.9 0.25% Casein hydrolysate BAP and NAA.

Define Growth promoting hormone:

Growth-promoting hormones, also known as plant growth regulators, are naturally occurring chemical substances that regulate and influence various aspects of plant growth and development. These hormones include auxins, cytokinins, gibberellins. They control processes such as cell division, elongation, differentiation, root and shoot development, flowering, fruiting, and response to environmental cues, playing critical roles in overall plant growth, morphogenesis, and physiological responses.

Disadvantages of PEG:

Some disadvantages of polyethylene glycol (PEG) in biological applications include its toxicity at high concentrations, potential interference with cellular processes, limited cell permeability, osmotic stress on cells, and variability in its effectiveness depending on the cell type and experimental conditions.

Enlist the method of chloroplast transmission:

The methods of chloroplast transmission in plants include maternal inheritance, somatic fusion, genetic engineering, chloroplast transplantation, and grafting, facilitating the transfer and study of chloroplasts between plant cells and tissues.